VIDEO GAME DATASET ANALYSIS

Dataset name: VideoGames.csv

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BUSAN 302 Group Project Report

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Executive Summary

This study aims to identify where future investment in video-game development is most likely to generate revenue. Using a processed dataset of 6,142 titles ranging from 1985-2016 that combines regional sales, critic and user ratings, genre, platform and release year. Three machine-learning approaches—linear regression, logistic regression and K-Means clustering was analysed to determine the most suitable model to answer our objective: "Which factors have the greatest impact on global sales?"

Key Model Findings

- K-Means clustering proved to be the most insightful model, with k = 2 the model reached a silhouette score of 0.536 and separated titles into a broad, moderate-performance cluster spanning multiple decades and a high-sales, high-critic-score cluster consisting of 1990s Sports, Platform, Racing, Action, RPG and Fighting games. These top performers averaged 1.23 million units in North America alone and succeeded from both nostalgia values and strong critic and user reception.

Strategic Implications

- Target nostalgia-rich genres from the 1990s. The best-performing cluster suggests that reviving or remastering familiar IPs in Sports, Platform and related genres can receive broad attention from two generations of gamers across the globe, much like the recent success of Dungeon Fighter Online's mobile relaunch.

Introduction

In today's competitive video game industry, understanding what makes a game a global bestseller is crucial for developers, publishers, and investors. Our project, called "Data-Driven Game Market Analysis," aims to answer this question by using data mining techniques and machine learning models to explore and predict global sales performance based on key game attributes.

We used a cleaned dataset containing 6,142 entries, each of which provided information such as critics' ratings, user ratings, platforms, genres, release years, and sales in different regions. After preprocessing, we asked two core questions:

1. Which factors have the greatest impact on global sales?

2. Can we use these variables to build an effective predictive model?

To answer these questions, we used a variety of machine learning methods: linear regression, K-Means clustering, and logistic regression. Each method helps us discover different aspects of the data. For example, linear regression can help us gain insight into the continuous relationship between numerical variables such as critics/user ratings and sales, while K-Means clustering can help us group games according to categorical patterns such as genre and decade.

Descriptive statistics

Our analysis involves using a video game dataset, which includes game attributes such as the platform, genre, release year, regional sales (North America, Europe, Japan, and other regions), global sales, critic scores, and user scores.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6250 entries, 0 to 6249
Data columns (total 11 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Name             6250 non-null   object
 1   Platform         6250 non-null   object
 2   Year_of_Release  6250 non-null   int64
 3   Genre            6250 non-null   object
 4   NA_Sales         6229 non-null   float64
 5   EU_Sales         6236 non-null   float64
 6   JP_Sales         6233 non-null   float64
 7   Other_Sales      6236 non-null   float64
 8   Global_Sales     6232 non-null   float64
 9   Critic_Score     6231 non-null   float64
 10  User_Score       6235 non-null   float64
dtypes: float64(7), int64(1), object(3)
memory usage: 537.2+ KB
```

Figure 1. Overview of the dataset structure showing columns, non-null counts, and data types.

The first steps of our analysis involved deciding on our target variable and what variables to include in our clustering model. Marchand and Hennig-Thurau (2013) indicated that game sales are significantly influenced by critic and user scores, as they reflect the game's reception and popularity. Additionally, Zhang (2021) highlight the importance of regional sales and genre as key determinants of a game’s market performance (pp. 222–227). Therefore, we have decided to include the listed variables below in our analysis to identify the relationships between these variables and game clustering patterns.

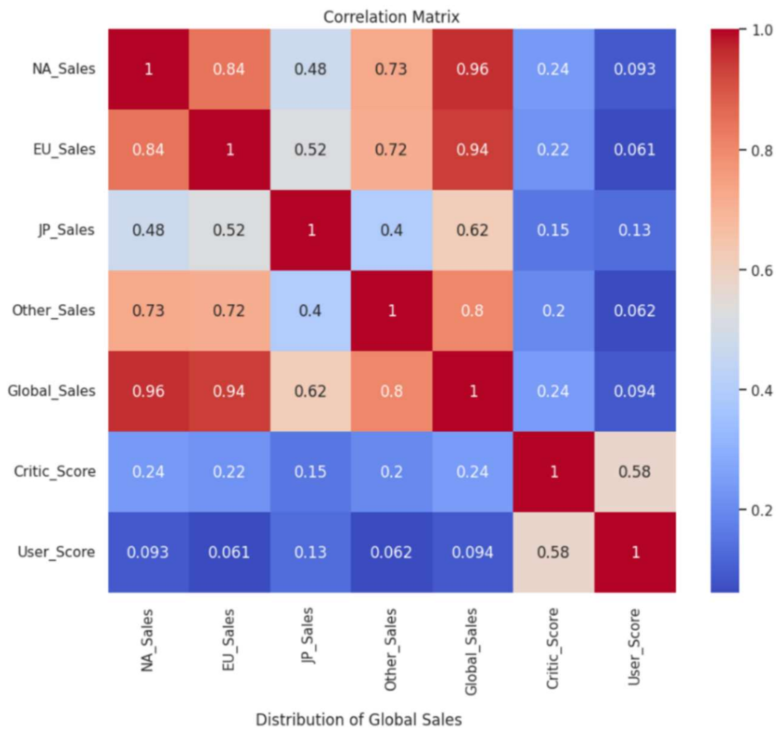


Figure 2. Heatmap of Correlation Matrix for Sales Data and Review Scores

In our analysis, we have focused on the numerical variables to identify their relationships with global sales. The correlation heatmap below illustrates the strength and direction of these relationships. Based on the dataset, we expect positive correlations between regional sales (NA_Sales, EU_Sales, JP_Sales, Other_Sales), critic scores, user scores, and global sales, indicating that higher values in these variables are associated with increased global sales. This assumption is supported by the clustering results, where games with higher sales and scores tend to group together.

The line chart (Figure 3) shows the average global video game sales figures grouped by decade from 1980 to 2010. In the 1980s, sales started off low, at around 100,000 units, and then surged to approximately 2 million units in the 1990s, likely due to the rise of popular gaming consoles such as the PlayStation and Nintendo 64. Sales then significantly declined, reaching about 750,000 units by the 2000s, and later stabilized at around 800,000 units in the 2010s. This trend highlights the 1990s as the peak period for game sales, which is a finding that intrigues us.

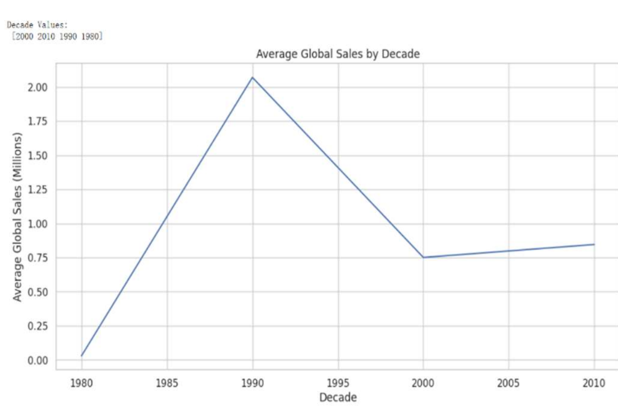


Figure 3. Average Global Sales by Decade

Data Pre-Processing

Data cleaning must be done before we start analyzing our dataset, as it is to ensure we get more accurate statistics and better predictions for real-life implementation later. The following steps are processed before training our model.

Handling Missing Values

Missing values generate errors when we attempt to use models, because most models require complete data and cannot process null values properly. Instead of filling the gaps with mean or median, we chose to drop all the missing values. It is because different games have too big differences in terms of their sales and quality across 1985 to 2016, and applying median or mean may result in values significantly different from how the game is in reality.

Drop Duplicates & Outliers

Duplicated values significantly reduce the accuracy of the model with distorted statistics. Counting the same value multiple times misleads the averages, totals and frequencies of the data. To ensure data quality, we drop duplicates before beginning our analysis. Additionally, sales should always be positive, as it would give an unrealistic performance for games in the model with unusual data. To ensure accurate results, we removed these outliers and selected only the records with sales greater than zero for our analysis.

Figure 4. Overview of steps for the removal of duplicates, outliers and missing values.

Standard Scaling

The variables in the dataset have significantly different ranges: sales range from 0 to 82, user scores range from 0.5 to 9.6, and critic scores range from 13 to 98. These differences in scale can negatively affect the performance of machine learning algorithms, especially K-means clustering, which rely on distance metrics. Standardisation was applied to ensure that each feature contributes equally to the analysis and to improve model accuracy. This process transforms the data to have a mean of 0 and a standard deviation of 1, meaning that all features are on the same scale without distorting their distributions. Models trained on this data are more stable and reliable.

Add the Decade Column

My group chose to categorize the games by the decade they were released, rather than using the year. It is because game trends often vary a lot from year to year. Grouping games by decades provides a clearer view of long-term industry trends. Additionally, using decades ensures that each cluster contains enough data, which is important for the effectiveness of K-means clustering. This helps improve the stability and accuracy of the model's predictions.

For detailed Code snippet of our data preprocessing please refer to APPENDIX 1.0.

Model Overview

After Data Preprocessing, Using the target of "Global Sales" and various features such as 'NA_Sales', 'EU_Sales', 'JP_Sales', 'Other_Sales', 'Critic_Score', 'User_Score'. We experimented with three modelling approaches—linear regression, logistic regression and K-means. For detailed model summary please see Appendix 2.0

linear regression

We first attempted to predict a continuous outcome global unit sale from genre, critic scores and user scores.

The simple linear model had a **R^2** of 0.185, the model cannot recognize why some games in certain genre sells higher than others.

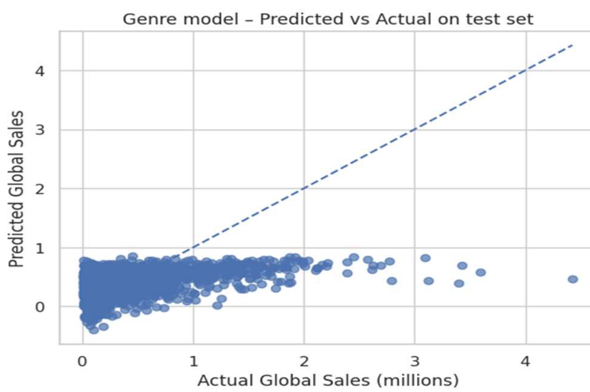


Figure 5. predicted vs actual global sales of linear regression

The target variable itself might be the problem, we tried to predict critic scores instead, now feeding the model with platform, year of release and global sales added. The model improved performance to an adjusted **R^2** of 0.46 and an **RMSE** of 10 rating score. A big gain in accuracy from previous but it very weakly relates to our overall question and another approach is attempted.

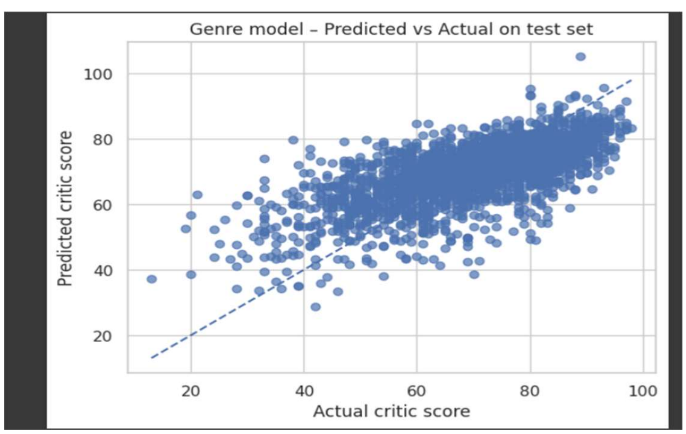


Figure 6. predicted vs actual Critic Score of linear regression

Logistic regression

We reframed the problem as a classification task. We labelled any title selling at least half a million copies worldwide as a success and everything else as a failure to investors. The model's overall **accuracy** was only 59 percent.

```
Classification report:
              precision    recall  f1-score   support

           0      0.799     0.542     0.646      1293
           1      0.398     0.690     0.505       567

    accuracy                          0.587      1860
   macro avg      0.599     0.616     0.575      1860
weighted avg      0.677     0.587     0.603      1860

ROC-AUC: 0.649
```

Figure 7. ROC curve of logistic regression with classification report

The **precision** stayed below 40 percent, the algorithm routinely called "hit" on games that failed to meet the requirement. This suggests that the model is almost always predicting a game as success and it is no more helpful than a coinflip, we attribute the reason to having too little information on the data variance again, making the model unsuitable for answering our question.

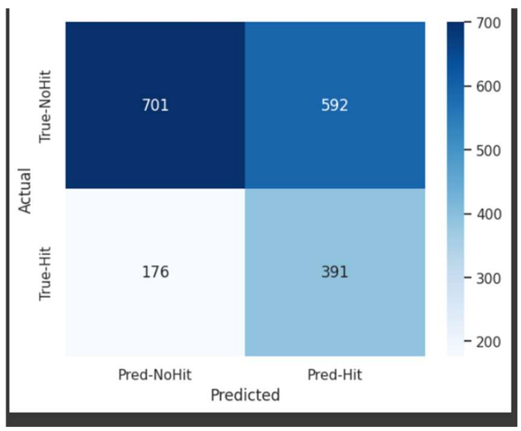


Figure 8. Confusion matrix of logistic regression

K-Means Clustering

We decided to shift to an unsupervised technique, K-means clustering to let the dataset structure reveal itself. Setting the genre as the goal for grouping, based on individual regional sales and critical scores, tested on four clusters, the solution produced a moderate **silhouette score** of 0.236, a reasonable separation with room for refinement. The model showed clear patterns that differ from regression models. One cluster captured shooters and strategy titles that sell disproportionately well in North America and Europe, while two further clusters skewed toward genres with stronger interests in Japan. A fourth, smaller group contained niche games that rarely become blockbusters but earn consistently high critic acclaim. Developers can research into the games within each genre to analyse what worked and what didn't, audience interests and other possibilities.

K-Means clustering is therefore chosen as our model of focus, to proceed, several enhancements could make the clustering even sharper. Such as adding in decade of release with the genre grouping will reveal genre popularity trends.

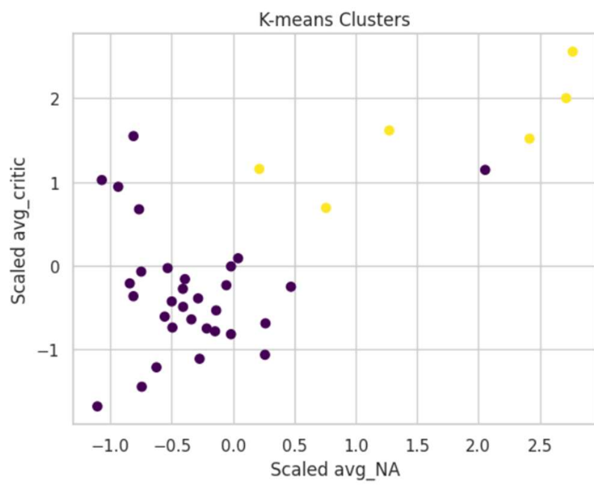
For detailed Clustering Table please see APPENDIX 2.1

Model results

To determine the optimal number of clusters for the VideoGames dataset, we applied K-means clustering with k=2 and k=3. We evaluated the clustering performance using multiple methods, including visual inspection of **scatter plots**, the **elbow method**, and **silhouette scores**.

A scatter plot of average North American sales (avg_NA) versus average critic score (avg_critic) for k=2 revealed clear separation, with 31 points in Cluster 0 (purple) and 6 points in Cluster 1 (yellow), indicating distinct groupings.

Figure 10. K-means Clusters of Scaled Average North American Sales vs. Scaled Average Critic Scores



To further guide the selection of k, we employed the **elbow method** (GeeksforGeeks, 2025), which involves calculating the Within-Cluster Sum of Squares (WCSS)—the sum of squared distances between each data point and its cluster centroid—for varying k. The WCSS is plotted against k to form an elbow plot, where the "elbow" point, where the WCSS reduction rate slows, suggests the optimal number of clusters. In our analysis, the elbow plot suggested that k=3 was also viable, as the WCSS began to plateau. **However, the silhouette score for k=2 was 0.536, higher than that for k=3(silhouette score = 0.471)**, indicating better cluster cohesion and separation. Thus, we selected k=2 as the optimal number of clusters.

```
#Elbow Method
wcss = []
k_range = range(1, 11)  # Test k from 1 to 10
for k in k_range:
        kmeans = KMeans(n_clusters=k, random_state=0, n_init=10)
        kmeans.fit(X_scaled)
        wcss.append(kmeans.inertia_)  # Inertia is WCSS

#Plot the Elbow Curve
plt.figure(figsize=(8, 5))
plt.plot(k_range, wcss, marker='o')
plt.title('Elbow Method for Optimal k')
plt.xlabel('Number of Clusters (k)')
plt.ylabel('Within-Cluster Sum of Squares (WCSS)')
plt.grid(True)
plt.show()
```

Figure 11. Code for Elbow Method

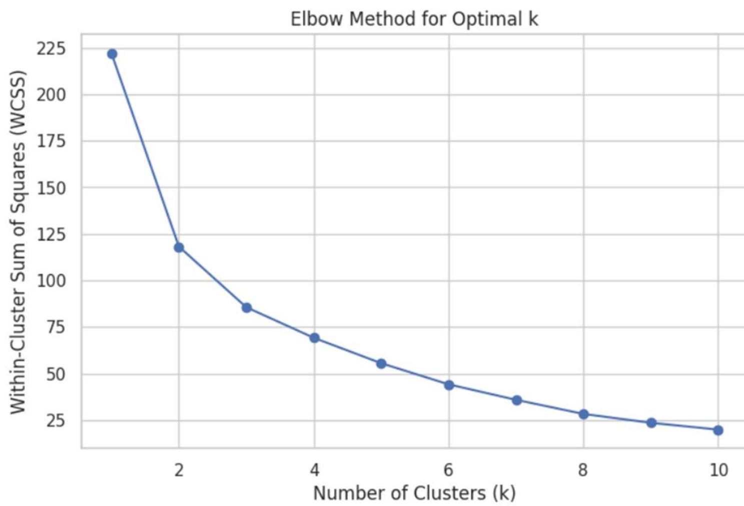


Figure 12. Elbow Method for Optimal k

The K-means clustering with k=2 identified two distinct clusters based on Genre-Decade pairs:

- **Cluster 0** (31 pairs): This cluster represents a diverse set of genres spanning from 1990 to 2010, with an average North American sales of 0.34 million units and an average critic score of 71.70. This group likely includes a broad range of games with moderate success across various platforms and time periods.
- **Cluster 1** (6 pairs): This cluster is characterized by significantly higher sales, averaging 1.23 million units in North America, and a higher average critic score of 85.81. It primarily consists of Sports and Platform genres from the 1990s, suggesting that these genres were particularly successful during that decade, possibly due to the popularity of gaming consoles like the PlayStation and Nintendo 64.

Figure 13. Genre × Decade Clusters with K-means (k=2)[Appendix 2.2]

These findings highlight a clear distinction between a smaller group of highly successful, critically acclaimed games from the 1990s (Cluster 1) and a larger, more varied group of games with lower sales and ratings across multiple decades (Cluster 0). The results underscore the influence of genre and time period on game performance, providing valuable insights for understanding historical trends in the video game industry.

```
Cluster-level means:
         pairs  mean_NA  mean_EU  mean_JP  mean_Other  mean_cscore  \
cluster
0           31     0.34      0.2      0.1        0.06        71.70
1            6     1.23      0.8      0.5        0.14        85.81

        mean_uscore
cluster
0              7.14
1              8.60
```

Figure 14. Cluster-Level Means for K-means Clustering (k=2)

Conclusion

Business Implication

Based on our analysis, we recommend focusing on Cluster 1, which demonstrated the highest sales performance. This cluster includes games released in the 1990s across genres such as Sports, Platform, Racing, Action, Role-Playing, and Fighting. Our primary target audience should be game publishers who are currently developing or planning to launch new titles. Given that the dataset is closely tied to game revenue, it has significant potential for generating insights that inform publishers' decision-making and help maximise profits.

From a geographical perspective, North America and Europe consistently showed the strongest sales performance, suggesting that these regions should be prioritised for marketing to generate more revenue for game publishers.

Our findings identified genre and release decade as key drivers of game sales. We notice that games released in the 1990s achieved the highest combined scores and sales. Therefore, we can develop a targeted strategy: focus on nostalgia by appealing to middle-aged players who are familiar with these games from their youth, especially within the Sports, Platform, Racing, Action, Role-Playing, and Fighting genres.

A real-world example that supports our strategy is Dungeon Fighter Online. Originally launched in 2005 as a browser game, its mobile version was released in China in 2024. It rapidly became one of the top-grossing mobile games, outperforming many newer titles (CyberEggTart9527, 2024). Even a year after its release, it consistently ranks among the top three in mobile game revenue in China.

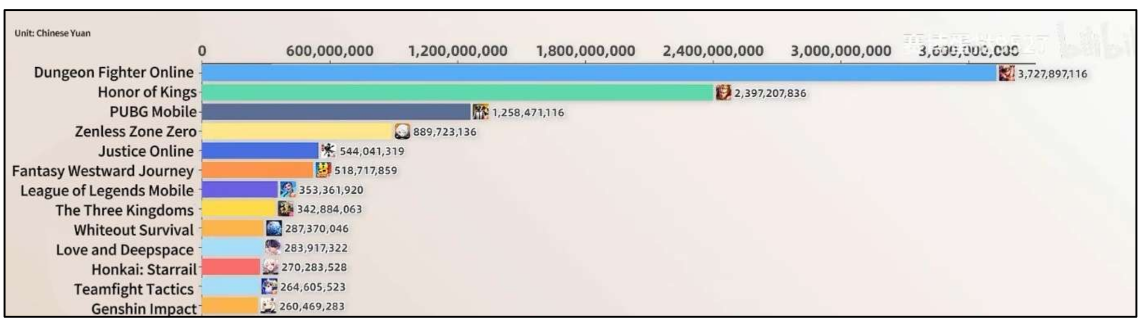


Figure 15. Top 13 mobile games’ monthly revenue in China for July 2024. (Chart created by CyberEggTart9527 using data from Sensor Tower, 2024.)

One major factor contributing to its success is the nostalgia effect. Many of its current players are middle-aged adults who played the original version in their youth. Because they are usually more financially stable, they are more willing to spend money on games they are emotionally connected to. This case strongly supports our conclusion that reviving or reinventing well-loved games from past decades can be a highly profitable business strategy.

Limitations

A major limitation of our analysis is the temporality of the dataset, which ends in 2016. This limits the model's ability to reflect current trends, such as mobile gaming, digital distribution, or platforms such as the Nintendo Switch and PS5. Outdated data with uneven temporal distribution can distort insights and reduce the model's relevance in today's market.

Biases in the dataset can also affect reliability. For example, average sales in the 1990s are skewed by a few blockbuster games, and user ratings often reflect nostalgia rather than objective quality. Harry Potter fans may rate Hogwarts Saga highly despite its flaws, illustrating the risks of subjective input (Joint Economic Committee, 2024).

Finally, models such as K-Means and linear regression can oversimplify the relationship between features and sales. More advanced methods such as random forests or XGBoost are better suited to capture complex patterns in the data (Han, Kamber, & Pei, 2012).

Ethics

While predicting game sales may not have life-changing consequences like they do in healthcare or finance, ethical considerations still apply. First, our model uses user-generated scores, which are inherently biased and can distort perceptions of game quality. Using these biases for business decisions without human oversight is particularly risky.

Our findings also highlight the importance of context in interpreting data. Emotional factors such as nostalgia can influence user ratings, raising ethical questions if models treat such inputs as objective. As the Joint Economic Committee (2024) noted in the housing sector, context-aware analyses are essential to achieving fair outcomes.

Finally, transparency and explainability are critical. Stakeholders, including developers and marketers, must understand the model's limitations and assumptions before relying on its predictions.

Final remarks

In summary, our project revealed major patterns between critic ratings, game genres, platforms, and global sales. K-Means clustering showed that games can be grouped by sales and rating trends, with classic games from the 1990s forming a high-performing cluster and modern mainstream games forming a more modest cluster.

However, these findings are limited by the age of the dataset and reliance on basic models. As shown in real estate forecasts, stronger forecasts require newer, more diverse data (Reserve Bank of New Zealand, n.d.). Future work should employ newer datasets and advanced models to improve the accuracy of forecasts.

This project highlights the potential of data-driven strategies in game development and marketing - if based on current, unbiased data and interpreted with human judgment and ethical care.

References

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Zhang, X. (2021). Evaluation of Game Publishers and Study on the Player Preference Based on the Data from 1980 to 2020 in Video Game Market. *2021 The 8th International Conference on Industrial Engineering and Applications(Europe)*, 222–227. https://doi.org/10.1145/3463858.3463870

Appendix 1.0

Data cleaning and preprocessing code

```
#Check for missing values
print(data.shape)
print('Missing Values:')
print(data.isnull().sum())

#Impute missing scores with median
#imputer = SimpleImputer(strategy='mean')
#data[['Critic_Score', 'User_Score']] = imputer.fit_transform(data[['Critic_Score', 'User_Score']])
#data = data[data['Global_Sales'] <= 10].copy()
#Drop rows with missing sales data
data = data.dropna(subset=['NA_Sales', 'EU_Sales', 'JP_Sales', 'Other_Sales', 'Global_Sales','Critic_Score','User_Score'])

#Remove duplicates
data = data.drop_duplicates()

#Ensure sales are non-negative
data = data[(data['NA_Sales'] >= 0) & (data['EU_Sales'] >= 0) & (data['JP_Sales'] >= 0) &
            (data['Other_Sales'] >= 0) & (data['Global_Sales'] > 0)]

#Verify cleaning
print('\nAfter Cleaning - Missing Values:')
print(data.isnull().sum())
print('\nShape after cleaning:', data.shape)
```

```
(6250, 11)
Missing Values:
Name              0
Platform          0
Year_of_Release   0
Genre             0
NA_Sales         21
EU_Sales         14
JP_Sales         17
Other_Sales      14
Global_Sales     18
Critic_Score     19
User_Score       15
dtype: int64

After Cleaning - Missing Values:
Name             0
Platform         0
Year_of_Release  0
Genre            0
NA_Sales         0
EU_Sales         0
JP_Sales         0
Other_Sales      0
Global_Sales     0
Critic_Score     0
User_Score       0
dtype: int64

Shape after cleaning: (6142, 11)
```

Appendix 2.0

ML model overview table

Algorithm	Task	Test Performance	Brief Description
Linear Regression	Global Sales prediction	$R^2 = 0.185$	Simple baseline, underfits, non-linear relationships
Linear Regression	Critic Score Prediction	$R^2 = 0.458$	Changing target and features offered marginal gain, although not the best for business question
Logistic Regression	High-sale vs Low-sales classification	Accuracy = 58.7%	Low precision with too high false flagging, indicating data isn't suitable
K-Means Clustering	Cluster by Genre k=4	Silhouette = 0.236	Clustering presents clear information regarding what genre of games is popular in each region
K-Means Clustering	Cluster by Genre X Decade K = 3	Silhouette = 0.471	Factoring in Decade of release increased clustering tightness marginally
K-Means Clustering	Cluster by Genre X Decade K = 2	Silhouette = 0.536	K=2 offered the best clustering, and information can be extracted comfortably

Appendix 2.1

Clustering Table of Initial K means grouped by genre only

Silhouette score: 0.236

Genre × Platform clusters:

— Cluster 0 —

	Genre	avg_NA	avg_EU	avg_JP	avg_Other	avg_critic	avg_user	n_games	cluster
6	Racing	0.411550	0.300523	0.054535	0.106143	69.153101	7.093411	516	0
8	Shooter	0.526008	0.309677	0.022351	0.104354	70.789406	7.087597	774	0
10	Sports	0.497699	0.272418	0.039042	0.103773	74.037383	7.058879	856	0
0	Action	0.366592	0.240971	0.046959	0.092376	67.380177	7.056687	1473	0
3	Misc	0.601647	0.323671	0.093555	0.108584	66.942197	6.789306	346	0

— Cluster 1 —

	Genre	avg_NA	avg_EU	avg_JP	avg_Other	avg_critic	avg_user	n_games	cluster
4	Platform	0.482715	0.270665	0.109169	0.082881	69.559557	7.303324	361	1
2	Fighting	0.358653	0.161138	0.075539	0.067275	68.940120	7.249401	334	1
5	Puzzle	0.297685	0.218704	0.137130	0.057130	71.027778	7.230556	108	1
9	Simulation	0.320746	0.232463	0.098172	0.059142	69.839552	7.150373	268	1

— Cluster 2 —

	Genre	avg_NA	avg_EU	avg_JP	avg_Other	avg_critic	avg_user	n_games	cluster
7	Role-Playing	0.31206	0.173043	0.183233	0.058716	72.527734	7.580507	631	2

— Cluster 3 —

	Genre	avg_NA	avg_EU	avg_JP	avg_Other	avg_critic	avg_user	n_games	cluster
11	Strategy	0.125732	0.092971	0.017448	0.023766	72.958159	7.255230	239	3
1	Adventure	0.151737	0.098390	0.031907	0.031483	66.016949	7.085593	236	3

```
Cluster-level means:
         pairs  mean_NA  mean_EU  mean_JP  mean_Other  mean_cscore  \
cluster
0            5     0.48     0.29     0.05        0.10        69.66
1            4     0.36     0.22     0.11        0.07        69.84
2            1     0.31     0.17     0.18        0.06        72.53
3            2     0.14     0.10     0.02        0.03        69.49

        mean_uscore
cluster
0              7.02
1              7.23
2              7.58
3              7.17
```

Appendix 2.2

Clustering table of Final k means grouped by genre and decade

Silhouette score: 0.536

Genre × Decade clusters:

—— Cluster 0 ——

	Genre	Decade	avg_NA	avg_EU	avg_JP	avg_Other	avg_critic	avg_user	n_games	cluster
0	Shooter	1990	1.395000	0.292500	0.022500	0.037500	82.500000	8.425000	4	0
1	Strategy	1990	0.016000	0.038000	0.000000	0.004000	81.600000	8.320000	5	0
2	Simulation	1990	0.130000	0.087500	0.455000	0.040000	85.500000	8.175000	4	0
3	Misc	1990	0.150000	0.095000	0.820000	0.025000	79.000000	8.150000	2	0
4	Puzzle	1990	0.075000	0.050000	0.000000	0.010000	81.000000	7.850000	2	0
5	Role-Playing	2000	0.307582	0.142343	0.188759	0.050707	71.936869	7.746582	397	0
6	Adventure	1990	0.270000	0.125000	0.250000	0.025000	68.500000	7.550000	2	0
7	Simulation	2000	0.362426	0.233134	0.082277	0.062129	71.089552	7.476733	202	0
8	Strategy	2000	0.116609	0.073506	0.021543	0.018391	72.419540	7.410857	175	0
9	Adventure	2010	0.129753	0.122469	0.023086	0.038500	71.283951	7.393827	81	0
10	Fighting	2000	0.393423	0.157568	0.070000	0.072072	68.417040	7.390135	223	0
11	Shooter	2000	0.427399	0.209615	0.011525	0.072293	70.021277	7.377606	520	0
12	Sports	2000	0.506246	0.247928	0.044403	0.099480	74.649371	7.375667	637	0
13	Platform	2000	0.480662	0.249596	0.090879	0.080294	67.908425	7.323810	273	0
14	Racing	2000	0.423342	0.267909	0.041914	0.107638	68.168342	7.272796	399	0
15	Puzzle	2000	0.308427	0.239091	0.153146	0.062135	70.337079	7.260674	89	0
16	Action	2000	0.367759	0.196410	0.041972	0.089402	65.726437	7.232221	871	0
17	Role-Playing	2010	0.315044	0.205771	0.125833	0.071096	72.798246	7.222026	228	0
18	Platform	2010	0.480805	0.332299	0.155517	0.090805	73.929412	7.154023	87	0
19	Puzzle	2010	0.252778	0.137778	0.070000	0.034444	73.777778	7.038889	18	0
20	Fighting	2010	0.242162	0.134286	0.052432	0.054107	69.468468	6.945946	112	0
21	Adventure	2000	0.159744	0.083924	0.032866	0.027658	63.242038	6.943038	158	0
22	Misc	2000	0.602983	0.335527	0.093025	0.116807	66.080169	6.830802	238	0

—— Cluster 1 ——

	Genre	Decade	avg_NA	avg_EU	avg_JP	avg_Other	avg_critic	avg_user	n_games	cluster
0	Sports	1990	1.710000	0.690000	0.365000	0.125000	93.000000	8.950000	2	1
1	Platform	1990	1.050000	0.578333	0.325000	0.098333	86.000000	8.783333	6	1
2	Racing	1990	1.686250	1.310000	0.682500	0.192500	88.875000	8.762500	8	1
3	Action	1990	1.554545	1.187273	0.420909	0.230909	85.272727	8.563636	11	1
4	Role-Playing	1990	0.582632	0.438421	0.700526	0.100526	82.578947	8.484211	19	1
5	Fighting	1990	0.822500	0.585000	0.530000	0.083750	79.125000	8.037500	8	1